

Discussion on dB

- ❑ RF Power is the measurement which is carried out by engineers so often. Power meter is usually used to measure integrated RF Power across the band of test. Spectrum Analyzer is used to measure instantaneous power at various frequencies of interest.
- ❑ It is measured usually in watts, but there are units such as dBm, dBW which help represent linear scale on decibel with respect to either 1 mW or 1 watt.
- ❑ **dB** is the ratio of either amplitude of voltage or current of the two signals. dB stands for Decibel and will be taken as log of ratio of two power values (or voltage).

Discussion on dB

- ❑ $\text{dB} = 10 \log(P_2/P_1)$ or $20 \log (A_2/A_1)$...Equation-1
- ❑ **dBm** stands for decibel with respect to 1 milliwatt.
- ❑ **dBW** stands for decibel with respect to 1 Watt
- ❑ Above explanation helps one understand difference between dB and dBm. Pls. note that unit decibel (dB) can be used for ratio of two similar quantities such as voltage. Hence both numerator and denominator should be voltage. dBm is specifically refers to power unit. It is expressed as follows:
 - ❑ $\text{Power (dBm)} = 10 * \log_{10}(\text{Power in milliwatt})$...Equation-2

Discussion on dB

Decibel (dB), is a specific function that operates on a **unitless** parameter:

$$dB \doteq 10 \log_{10}(x)$$

where x is unitless!

Q: *A unitless parameter! What good is that !?*

A: **Many** values are unitless, such as **ratios** and **coefficients**.

For example, amplifier **gain** is a unitless value!

E.G., amplifier gain is the ratio of the output power to the input power:

$$\frac{P_{out}}{P_{in}} = G$$

$$\therefore \text{Gain in dB} = 10 \log_{10} G \doteq G(dB)$$

Q: *Wait a minute! I've seen statements such as:*

.... the output power is 5 dBw
or
.... the input power is 17 dBm

*Of course, Power is **not** a unitless parameter!?!*

A: True! But look at how power is expressed; not in dB, but in dBm or dBw.

Q: *What the heck does dBm or dBw refer to ??*

A: It's sort of a trick !

Say we have some power P . Now say we divide this value P by one 1 Watt. The result is a unitless value that expresses the value of P in relation to 1.0 Watt of power.

For example, if $P = 2500 \text{ mW}$, then $P/1W = 2.5$. This simply means that power P is 2.5 times larger than one Watt!

Since the value $P/1W$ is unitless, we can express this value in decibels!

Specifically, we define this operation as:

$$P(\text{dBw}) \doteq 10 \log_{10} \left(\frac{P}{1 \text{ W}} \right)$$

For example, $P = 100$ Watts can alternatively be expressed as $P(\text{dBw}) = +20 \text{ dBw}$. Likewise, $P = 1 \text{ mW}$ can be expressed as $P(\text{dBw}) = -30 \text{ dBw}$.

Q: *OK, so what does dBm mean?*

A: This notation simply means that we have normalized some power P to one **Milliwatt** (i.e., $P/1 \text{ mW}$)—as opposed to one Watt. Therefore:

$$P(\text{dBm}) \doteq 10 \log_{10} \left(\frac{P}{1 \text{ mW}} \right)$$

For example, $P = 100$ Watts can alternatively be expressed as $P(\text{dBm}) = +50 \text{ dBm}$. Likewise, $P = 1 \text{ mW}$ can be expressed as $P(\text{dBm}) = 0 \text{ dBm}$.

Make sure you are very **careful** when doing math with decibels!

Standard dB Values

Note that $10 \log_{10}(10) = 10 \text{ dB}$

Therefore an amplifier with a gain $G = 10$ is likewise said to have a gain of **10 dB**.

Now consider an amplifier with a gain of **20 dB**.....

Recall from **your** knowledge of logarithms that:

$$10 \log_{10} [10^n] = n 10 \log_{10} [10] = 10n$$

Therefore, if we express gain as $G = 10^n$, we conclude:

$$G = 10^n \leftrightarrow G(\text{dB}) = 10n$$

In other words, $G = 100 = 10^2$ ($n=2$) is expressed as 20 dB, while 30 dB ($n=3$) indicates $G = 1000 = 10^3$.

Likewise 100 mW is denoted as 20 dBm, and 1000 Watts is denoted as 30 dBW.

Note also that 0.001 mW = 10^{-3} mW is denoted as -30 dBm.

Another important relationship to keep in mind when using decibels is $10\log_{10}[2] \approx 3.0$. This means that:

$$10\log_{10}[2^n] = n \cdot 10\log_{10}[2] \simeq 3n$$

Therefore, if we express gain as $G = 2^n$, we conclude:

$$G = 2^n \leftrightarrow G(\text{dB}) \simeq 3n$$

As a result, a 15 dB ($n=5$) gain amplifier has $G = 2^5 = 32$. Similarly, $1/8 = 2^{-3}$ mW ($n=-3$) is denoted as -9 dBm.

Multiplicative Products and Decibels

Other logarithmic relationship that we will find useful are:

$$10\log_{10}[x y] = 10\log_{10}[x] + 10\log_{10}[y]$$

and its close cousin:

$$10\log_{10}\left[\frac{x}{y}\right] = 10\log_{10}[x] - 10\log_{10}[y]$$

Thus, the relationship $P_{out} = G P_{in}$ is written in decibels as:

$$\begin{aligned} P_{out} &= G P_{in} \\ \frac{P_{out}}{1mW} &= \frac{G P_{in}}{1mW} \\ 10\log_{10}\left[\frac{P_{out}}{1mW}\right] &= 10\log_{10}\left[\frac{G P_{in}}{1mW}\right] \\ 10\log_{10}\left[\frac{P_{out}}{1mW}\right] &= 10\log_{10}[G] + 10\log_{10}\left[\frac{P_{in}}{1mW}\right] \\ P_{out}(dBm) &= G(dB) + P_{in}(dBm) \end{aligned}$$

It is evident that "deebees" are **not** a unit! The units of the result can be found by **multiplying** the units of each term in a **summation** of decibel values.

For example, say some power $P_1 = 6 \text{ dBm}$ is combined with power $P_2 = 10 \text{ dBm}$. What is the resulting total power $P_T = P_1 + P_2$?

Logarithms are very helpful in expressing **products** or **ratios** of parameters, but they are **not** much help when our math involves sums and differences!

$$10\log_{10}[x + y] = \text{????}$$

So, if you wish to add $P_1 = 6 \text{ dBm}$ of power to $P_2 = 10 \text{ dBm}$ of power, you must first **explicitly** express power in Watts:

$$P_1 = 10 \text{ dBm} = 10 \text{ mW} \quad \text{and} \quad P_2 = 6 \text{ dBm} = 4 \text{ mW}$$

Thus, the total power P_T is:

$$\begin{aligned}P_T &= P_1 + P_2 \\&= 4.0 \text{ mW} + 10.0 \text{ mW} \\&= 14.0 \text{ mW}\end{aligned}$$

Now, we can express this total power in dBm , where we find:

$$P_T (\text{dBm}) = 10 \log_{10} \left(\frac{14.0 \text{ mW}}{1.0 \text{ mW}} \right) = 11.46 \text{ dBm}$$

The result is **not** 16.0 dBm !

We **can** mathematically add 6 dBm and 10 dBm , but we must understand what result means (nothing useful!).

$$\begin{aligned}6 \text{ dBm} + 10 \text{ dBm} &= 10 \log_{10} \left[\frac{4 \text{ mW}}{1 \text{ mW}} \right] + 10 \log_{10} \left[\frac{10 \text{ mW}}{1 \text{ mW}} \right] \\&= 10 \log_{10} \left[\frac{40 \text{ mW}^2}{1 \text{ mW}^2} \right] \\&= 16 \text{ dB relative to } 1 \text{ mW}^2\end{aligned}$$

Thus, mathematically speaking, $6 \text{ dBm} + 10 \text{ dBm}$ implies a multiplication of power, resulting in a value with units of **Watts squared**!

A few relations

1. $1.0 \leftrightarrow 0 \text{ dB}$

2. $0.0 \leftrightarrow -\infty \text{ dB}$

3. $5^n \leftrightarrow \approx 7n \text{ dB}$

$$X_{\text{dBW}} = X_{\text{dBm}} - 30.$$

Hence as per above equation,

$$0 \text{ dBW} = +30 \text{ dBm} \text{ or } -30 \text{ dBW} = 0 \text{ dBm}$$

dBm value	mw/Watt value
0 dBm	1 mw
30 dBm	1 Watt
40 dBm	10 watt
50 dBm	100 watt